

Advanced Algorithmic Forecasting Techniques

INTRODUCTION

This chapter introduces readers to advanced algorithmic forecasting techniques. We begin by reformulating our transaction cost equations in terms of the various trading strategy definitions, such as percentage of volume, trade rate, and trade schedules, and calibrate the parameters for these model variations. Estimated market impact costs for each approach are compared for the different data samples.

Readers are next introduced to the various sources of algorithmic trading risk including price volatility, liquidity risk, and parameter estimation uncertainty. We derive algorithmic forecasting techniques to estimate daily volume and monthly ADVs. The daily volume forecasting model is based on an autoregressive moving average (ARMA) time series which incorporates a median metric with a day of week effect adjustment factor, and the monthly ADV model incorporates previous volume levels, momentum, and market volatility.

The chapter concludes with an overview of the various transaction equations that are utilized to construct the efficient trading frontier and to develop optimal “best execution” strategies. All of which are essential building blocks for traders and portfolio managers interested in improving portfolio returns through best in class transaction costs management practices¹.

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